

Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- [jqassistant](#)
- [Neo4j Python Driver](#)

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Update, Change]	26416	48.602602
1	[Git, Commit]	6744	12.408235
2	[Git, Change, Create]	5847	10.757852
3	[Git, Change, Delete]	4190	7.709150
4	[File, Git]	3576	6.579456
5	[Git, Change, Rename]	1699	3.125977
6	[Author, Git, Person]	1000	1.839893
7	[Git, Tag]	906	1.666943
8	[Json, Key]	668	1.229048
9	[Json, Value, Scalar]	603	1.109455
10	[Committer, Git, Person]	356	0.655002
11	[NPM, Dependency]	338	0.621884
12	[Type, TS, Primitive]	294	0.540928
13	[Type, TS, Declared]	294	0.540928
14	[TS, ExternalDeclaration]	215	0.395577
15	[Type, TS, Literal]	138	0.253905
16	[Json, Value, Object]	133	0.244706
17	[Type, TS, Union]	121	0.222627
18	[Type, TS, ObjectMember]	111	0.204228
19	[NPM, Script]	91	0.167430
20	[TS, Property]	65	0.119593
21	[TS, Function]	47	0.086475
22	[TS, Parameter]	44	0.080955
23	[Type, Object, TS]	42	0.077275
24	[Type, TS, FunctionParameter]	40	0.073596
25	[File, Directory]	34	0.062556
26	[Type, TS, Function]	34	0.062556
27	[Package, File, Json, NPM]	29	0.053357
28	[TS, Variable]	24	0.044157
29	[Value, TS, Literal]	20	0.036798

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

<Figure size 640x480 with 0 Axes>

Nodes per label combination (more than 0.5% overall)

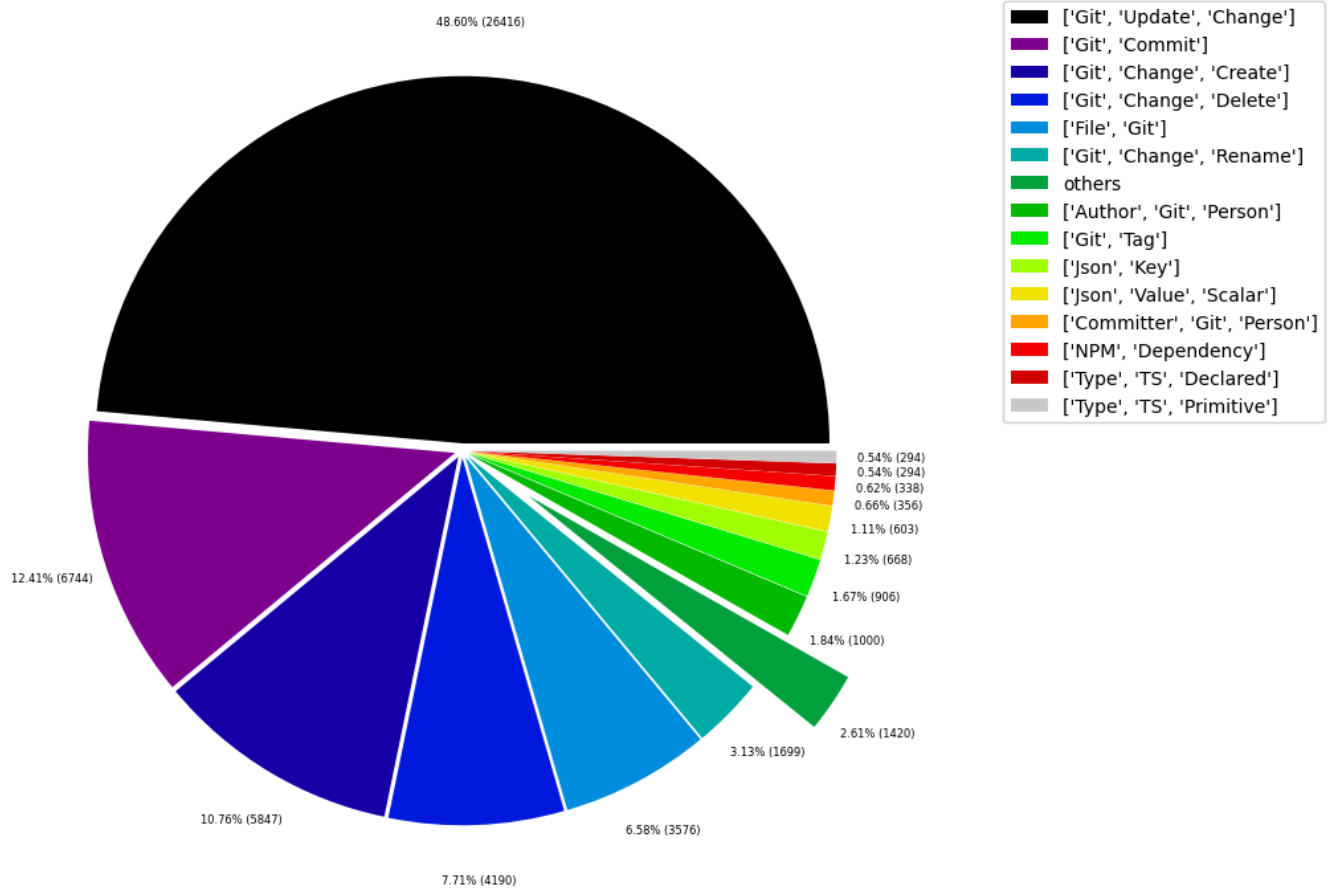


Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, JQAssistant]	1	0.001840
1	[NPM, PublishConfig]	1	0.001840
2	[File, TS, Scan]	1	0.001840
3	[TS, Method]	1	0.001840
4	[Git, Branch]	1	0.001840
5	[TS, Constructor]	1	0.001840
6	[Value, TS, ObjectMember]	1	0.001840
7	[Repository, File, Git]	1	0.001840
8	[TS, Class]	1	0.001840
9	[TS, Enum]	2	0.003680
10	[Value, Object, TS]	3	0.005520
11	[Type, TS, Tuple]	3	0.005520
12	[Value, TS, Function]	4	0.007360
13	[TS, TypeParameter]	4	0.007360
14	[Value, TS, Complex]	5	0.009199
15	[Repository, NPM]	5	0.009199
16	[NPM, Engine]	6	0.011039
17	[Project, TS]	6	0.011039
18	[File, Local]	6	0.011039
19	[File, TS, Local, Module]	6	0.011039
20	[Value, TS, Call]	6	0.011039
21	[Type, TS, TypeParameterReference]	6	0.011039
22	[Value, TS, Member]	6	0.011039
23	[TS, EnumMember]	8	0.014719
24	[Type, TS, NotIdentified]	11	0.020239
25	[TS, ExternalModule]	11	0.020239
26	[Json, Value, Array]	12	0.022079
27	[Value, TS, Declared]	13	0.023919
28	[Git, Change, Copy]	15	0.027598
29	[TS, TypeAlias]	16	0.029438

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Nodes per label combination (less than 0.5% overall)

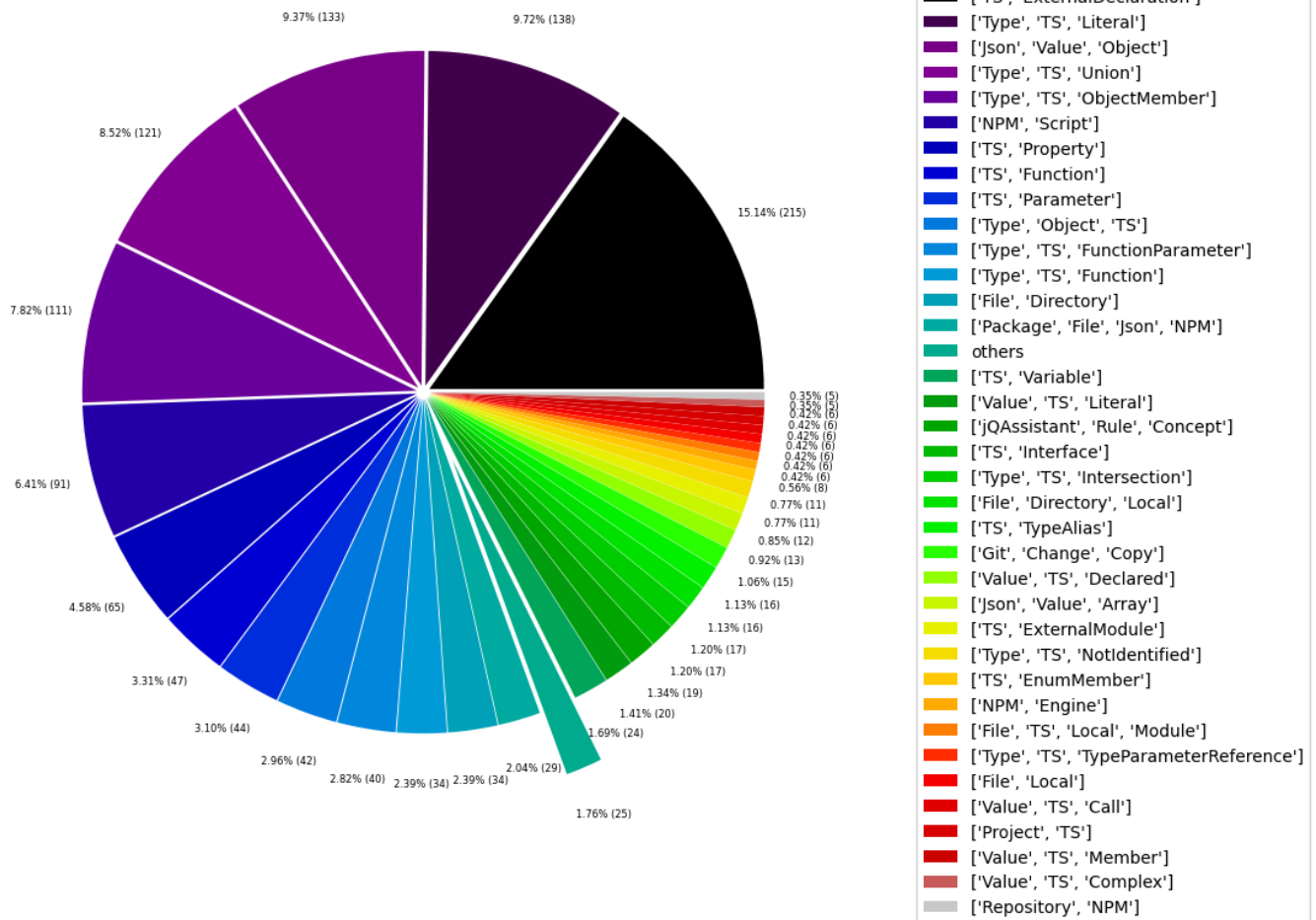


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

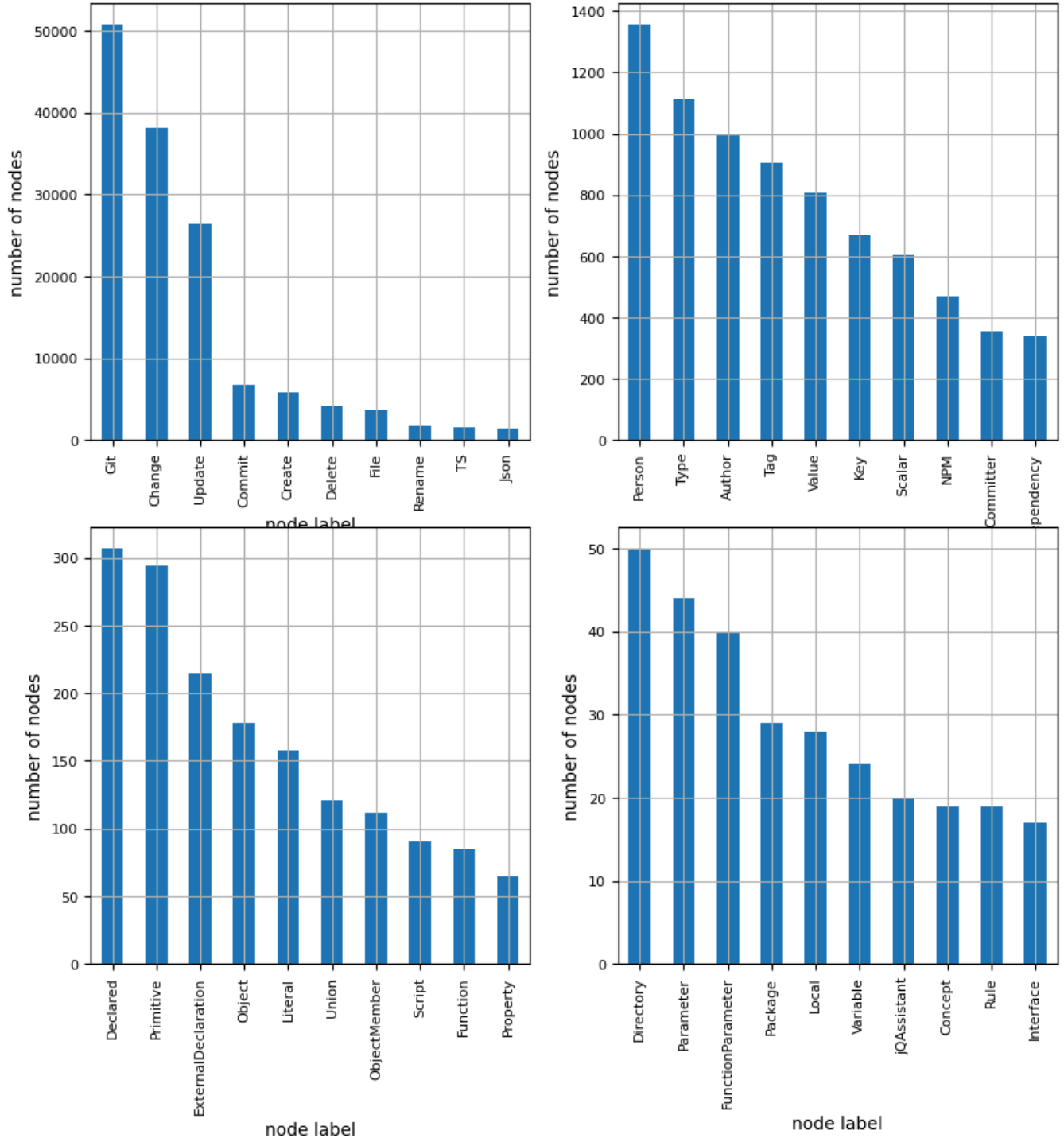
	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	50751	93.376387
1	Change	38167	70.223179
2	Update	26416	48.602602
3	Commit	6744	12.408235
4	Create	5847	10.757852
5	Delete	4190	7.709150
6	File	3669	6.750566
7	Rename	1699	3.125977
8	TS	1638	3.013744
9	Json	1445	2.658645
10	Person	1356	2.494894
11	Type	1111	2.044121
12	Author	1000	1.839893
13	Tag	906	1.666943
14	Value	806	1.482953
15	Key	668	1.229048
16	Scalar	603	1.109455
17	NPM	470	0.864749
18	Committer	356	0.655002
19	Dependency	338	0.621884
20	Declared	307	0.564847
21	Primitive	294	0.540928
22	ExternalDeclaration	215	0.395577
23	Object	178	0.327501
24	Literal	158	0.290703
25	Union	121	0.222627
26	ObjectMember	112	0.206068
27	Script	91	0.167430
28	Function	85	0.156391
29	Property	65	0.119593
30	Directory	50	0.091995
31	Parameter	44	0.080955
32	FunctionParameter	40	0.073596
33	Package	29	0.053357
34	Local	28	0.051517
35	Variable	24	0.044157
36	jQAssistant	20	0.036798
37	Concept	19	0.034958
38	Rule	19	0.034958
39	Interface	17	0.031278

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

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Node count by label



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

Total number of relationships: 160235

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	38167	23.819390
1	MODIFIES	38167	23.819390
2	UPDATES	26416	16.485787
3	COMMITTED	13488	8.417637
4	CREATES	7561	4.718694
5	HAS_PARENT	7517	4.691235
6	HAS_COMMIT	6744	4.208818
7	DELETES	5889	3.675227
8	HAS_FILE	3576	2.231722
9	CHANGED_TOGETHER_WITH	1801	1.123974
10	RENAMES	1699	1.060318
11	HAS_NEW_NAME	1032	0.644054
12	HAS_AUTHOR	1000	0.624083
13	HAS_TAG	906	0.565420
14	ON_COMMIT	906	0.565420
15	DEPENDS_ON	887	0.553562
16	HAS_KEY	668	0.416888
17	HAS_VALUE	668	0.416888
18	CONTAINS	599	0.373826
19	OF_TYPE	358	0.223422
20	HAS_COMMITTER	356	0.222174
21	EXPORTS	309	0.192842
22	REFERENCES	212	0.132306
23	DECLARES	186	0.116080
24	DECLARES_DEV_DEPENDENCY	169	0.105470
25	DECLARES_DEPENDENCY	161	0.100477
26	HAS_MEMBER	112	0.069897
27	HAS_TYPE_ARGUMENT	96	0.059912
28	DECLARES_SCRIPT	91	0.056792
29	HAS_PARAMETER	84	0.052423

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

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Relationship types (more than 0.5% overall)

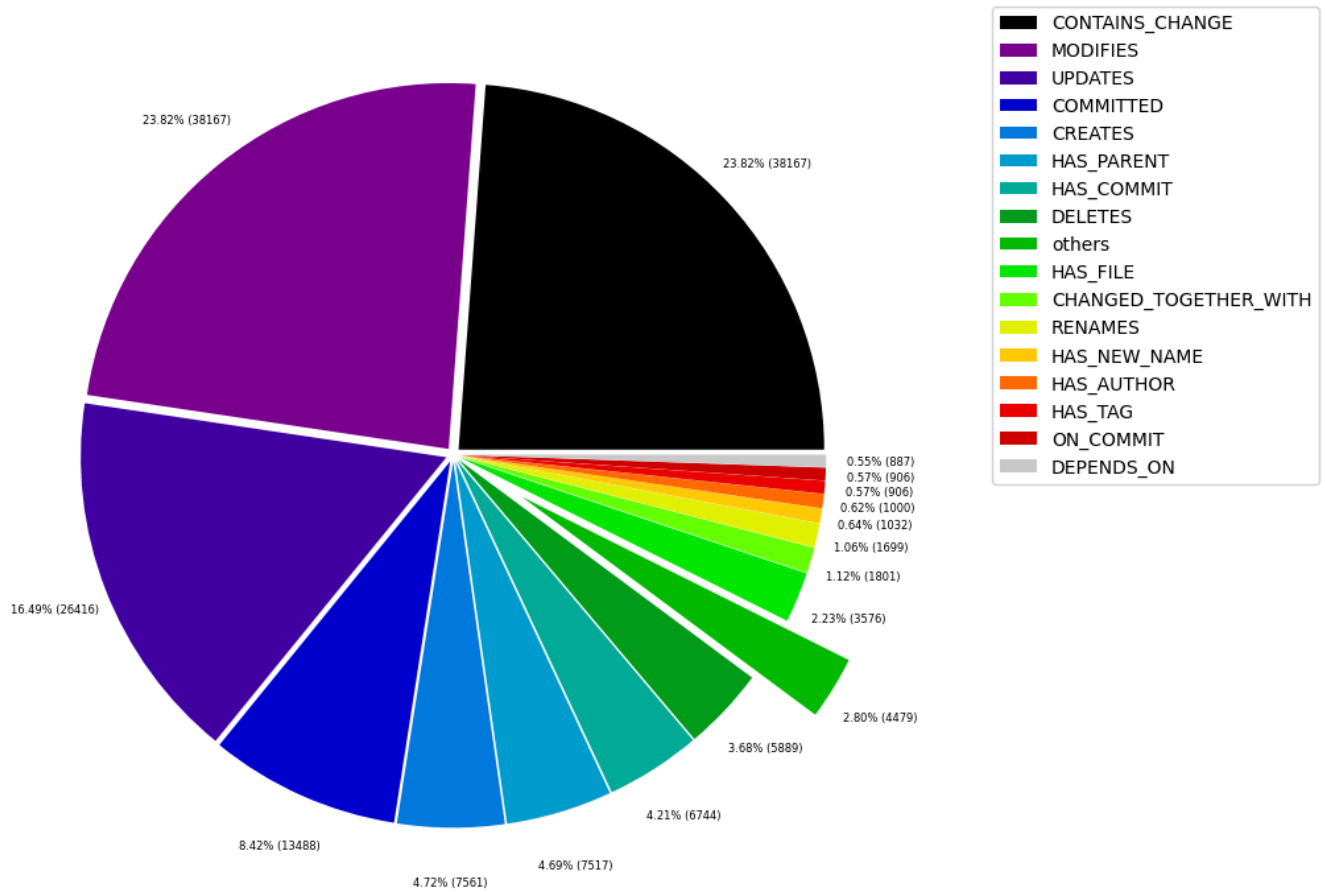


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

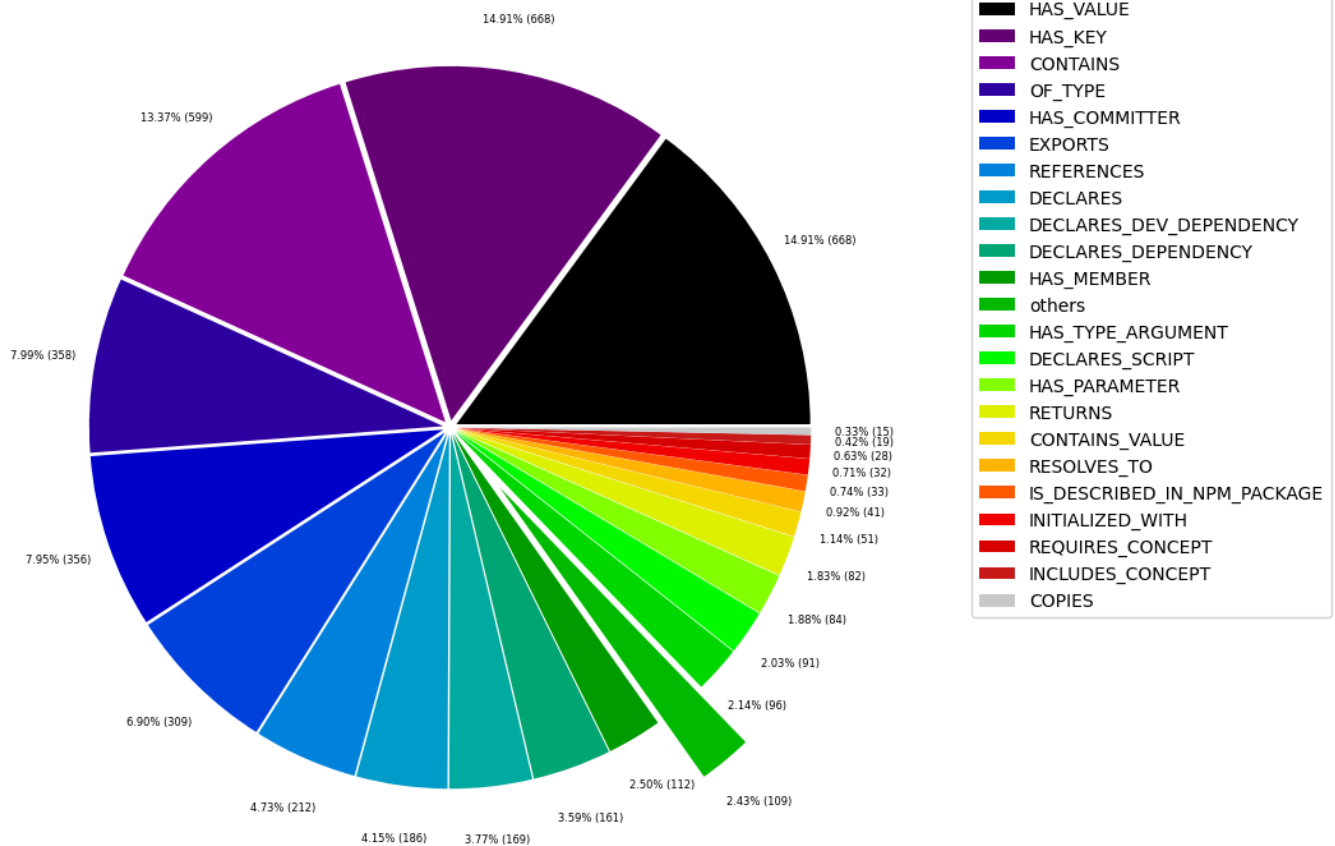
	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	HAS_BRANCH	1	0.000624
1	DECLARES_PUBLISH_CONFIG	1	0.000624
2	HAS_HEAD	2	0.001248
3	CONSTRAINED_BY	4	0.002496
4	REFERENCED_PROJECTS	5	0.003120
5	IN_REPOSITORY	5	0.003120
6	DECLARES_ENGINE	6	0.003745
7	EXTENDS	6	0.003745
8	HAS_ARGUMENT	6	0.003745
9	CALLS	6	0.003745
10	HAS_NPM_PACKAGE	6	0.003745
11	HAS_CONFIG	6	0.003745
12	PARENT	6	0.003745
13	CONTAINS_PROJECT	6	0.003745
14	MEMBER	6	0.003745
15	HAS_ROOT	6	0.003745
16	DECLARES_PEER_DEPENDENCY	8	0.004993
17	USES	11	0.006865
18	COPY_OF	12	0.007489
19	COPIES	15	0.009361
20	INCLUDES_CONCEPT	19	0.011858
21	REQUIRES_CONCEPT	28	0.017474
22	INITIALIZED_WITH	32	0.019971
23	IS_DESCRIBED_IN_NPM_PACKAGE	33	0.020595
24	RESOLVES_TO	41	0.025587
25	CONTAINS_VALUE	51	0.031828
26	RETURNS	82	0.051175
27	HAS_PARAMETER	84	0.052423
28	DECLARES_SCRIPT	91	0.056792
29	HAS_TYPE_ARGUMENT	96	0.059912

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Relationship types (less than 0.5% overall)



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numberOfNodesWithSameLabelsAsTarget
0	[Git, Update, Change]	MODIFIES	[File, Git]	26416	26416	
1	[Git, Update, Change]	UPDATES	[File, Git]	26416	26416	
2	[Git, Commit]	CONTAINS_CHANGE	[Git, Update, Change]	26416	6744	
3	[Git, Commit]	HAS_PARENT	[Git, Commit]	7517	6744	
4	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	6744	1	
5	[Author, Git, Person]	COMMITTED	[Git, Commit]	6744	1000	
6	[Committer, Git, Person]	COMMITTED	[Git, Commit]	6744	356	
7	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Create]	5847	6744	
8	[Git, Change, Create]	CREATES	[File, Git]	5847	5847	
9	[Git, Change, Create]	MODIFIES	[File, Git]	5847	5847	
10	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Delete]	4190	6744	
11	[Git, Change, Delete]	DELETES	[File, Git]	4190	4190	
12	[Git, Change, Delete]	MODIFIES	[File, Git]	4190	4190	
13	[Repository, File, Git]	HAS_FILE	[File, Git]	3576	1	
14	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Rename]	1699	6744	
15	[Git, Change, Rename]	RENAMES	[File, Git]	1699	1699	
16	[Git, Change, Rename]	CREATES	[File, Git]	1699	1699	
17	[Git, Change, Rename]	DELETES	[File, Git]	1699	1699	
18	[Git, Change, Rename]	MODIFIES	[File, Git]	1699	1699	
19	[File, Git]	CHANGED_TOGETHER_WITH	[File, Git]	1648	3576	
20	[File, Git]	HAS_NEW_NAME	[File, Git]	1032	3576	
21	[Repository, File, Git]	HAS_AUTHOR	[Author, Git, Person]	1000	1	
22	[Repository, File, Git]	HAS_TAG	[Git, Tag]	906	1	
23	[Git, Tag]	ON_COMMIT	[Git, Commit]	906	906	
24	[Json, Value, Object]	HAS_KEY	[Json, Key]	668	133	
25	[Json, Key]	HAS_VALUE	[Json, Value, Scalar]	552	668	
26	[Repository, File, Git]	HAS_COMMITTER	[Committer, Git, Person]	356	1	
27	[TS, Function]	DEPENDS_ON	[TS, ExternalDeclaration]	293	47	
28	[File, TS, Local, Module]	DEPENDS_ON	[TS, ExternalDeclaration]	236	6	
29	[TS, ExternalModule]	EXPORTS	[TS, ExternalDeclaration]	215	11	

Graph Density

```

total_number_of_nodes (vertices): 54351
total_number_of_relationships (edges): 160235
-> total directed graph density: 5.424382388101653e-05
-> total directed graph density in percent: 0.005424382388101653

```